Class 20: The Registrar's Report -- Instructor Notes

Survivorship bias in graduation statistics

Overview

30 minutes, quiz day. Files in `Class20/`, data = `registrar_report.csv` (63 rows, graduates only) and `full_cohort.csv` (100 rows, the entire entering cohort). Both generated from the same synthetic-cohort logic (seed 42): 60% graduate / 20% drop out / 20% still enrolled at the 6-year tracking window, with graduation times skewed around 4 years.

Placement: immediately follows the lecture's WWII bomber example (Abraham Wald, survivorship bias). Nothing new is introduced from scratch – the activity is a second, close-to-home instance of the same fallacy, worked through with a dataset instead of a story.

Arc: naive graduates-only average looks reassuring – commit to a prediction – reveal the 37 students the average silently dropped – a hand-built Kaplan-Meier curve shows the honest picture, including the part that's still unknown – close by mapping both cases onto the same table.

Timing

Part	Min	Content
1	5	Registrar's report (graduates only) – commit to a prediction on paper
2	12	Reveal full cohort – breakdown, bar chart, honest completion rate
3	10	Instructor-led Kaplan-Meier curve – the plateau, the shifted median
4	3	Paper mapping back to the bomber example

Part 1 – Key numbers

> **Note:** Registrar's report, 63 graduates: mean 4.31 years, median 4.00 years. Verified in the real `ds313` environment, not just a sandbox – exact match on both runs.

Enforce a real commit before anyone runs the naive-stat cell. The number by itself is not dramatic – 4.31 years sounds perfectly ordinary – which is exactly the point: nothing about it **looks** wrong. Poll a few predictions out loud before moving on; the spread (some students will say "accurate," some "too optimistic") sets up Part 2.

Part 2 – Key numbers

Status	Count	% of cohort
Graduate	63	63%
Drop out	19	19%
Still enrolled	18	18%

Honest completion rate = 63.0% exactly. The naive average was computed from just under two-thirds of the entering cohort, with no indication in the number itself that a third is missing.

> **Note:** If a team asks why "still enrolled" isn't just "will graduate eventually, ignore it" – that's the right question and the bridge into Part 3. We don't know yet. Some of them will graduate late, some never will, and today's snapshot can't tell the difference.

Part 3 – Key numbers (Kaplan-Meier)

Year	% not yet graduated
0.0	100.0%
3.5	96.4%
4.0	50.6%
4.5	39.2%
5.0	30.4%
5.5	24.0%
6.0	22.8%

The 50%-graduated point falls at **year 4.5** – half a year later than the naive graduates-only mean of 4.31. Direction matters for the debrief: the naive number isn't just incomplete, it's also a little too optimistic, because it's silently built from the students who succeeded fastest and never has to answer for the ones still working on it.

> **Note:** The notebook now plots censoring explicitly, matching the lifelines-style convention from your original prototype: 37 tick marks (|) on the curve, one per dropout or still-enrolled student, placed at their own tracked year and the survival level in effect at that point. All 18 still-enrolled students land on the same point (year 6, 22.8%) since they share a censoring time – worth pointing out that the stacked tick is real, not a rendering glitch. The notebook also prints the 19/18 dropout/still-enrolled split right before the plot.

Q3.1 target answer: 22.8% of the entering cohort has still not graduated at year 6, and no – this is not recoverable from the graduates-only report, which by construction only ever contains people who already crossed the finish line.

Q3.2 target answer: the naive average is biased **low** (too fast/optimistic) relative to the honest curve, because it's an average over survivors of the process, computed at the moment they succeeded – structurally similar to why the returning bombers only show survivable damage.

> **Caution:** Simplification worth naming if a sharp student pushes on it: this treats drop-out and still-enrolled identically as "censored." That's a simplification – drop-out is closer to a permanent competing outcome than to temporary censoring – but distinguishing them properly is more machinery than 30 minutes affords. One sentence of honesty is enough; don't let it eat the clock.

Part 4 – Answer key

What we can see	The 63 students who graduated, and how long each took.
What's missing from the data	The 19 dropouts and 18 still-enrolled – 37 students entirely absent from the report.
The naive conclusion	"Students graduate in about 4.3 years" – with the 100% completion rate this implies never stated, and never checked.
Why it's wrong	The missing 37 aren't a random subset – struggling is correlated with dropping out or taking longer, so excluding them doesn't just shrink the sample, it tilts it toward the easy cases. Same shape as the bombers: the missing cases are missing **for a reason connected to the very thing being measured**.

Q4.2, pocket examples if a team is stuck: startup-founder success books (built from companies that survived); published research (file-drawer problem – failed studies don't get written up); long-term-smoker health surveys (early deaths aren't in the "long-term" sample); investment fund track records (funds that failed quietly close and vanish from the dataset). If the Silberzahn/replication-crisis thread has come up before, publication bias is the cleanest callback.

Open items

- **Font fallback** – same Liberation Serif issue as Classes 16–19. Unresolved; left unchanged so this handout matches the rest of the series.
- **`preamble.typ` is still the reconstruction**, not your real preamble – same open item carried from every class since 16. Paste yours over it before compiling for real.
- **Data files are new, not reused** – `registrar_report.csv` and `full_cohort.csv` were generated fresh for this class (seed 42) and verified end-to-end in `ds313`: CSV generation, every notebook cell (with blanks filled), and the Kaplan-Meier function all ran without error, and the printed numbers match this document exactly.
- **Notebook code deliberately kept simple.** `kaplan_meier` uses `np.unique` instead of `sorted(set(...))`, builds arrays with `make_array`/`np.append` (the same pattern used for bootstrap loops in Class 19) instead of Python lists, returns a single `Table` instead of a tuple to unpack, and every loop is written with `np.arange` indexing instead of `zip`. More lines than the terse version, on purpose – nothing here should require a Python idiom students haven't already seen in a lab.